MULTI-MODAL ONCOLOGY AI ANALYSIS REPORT

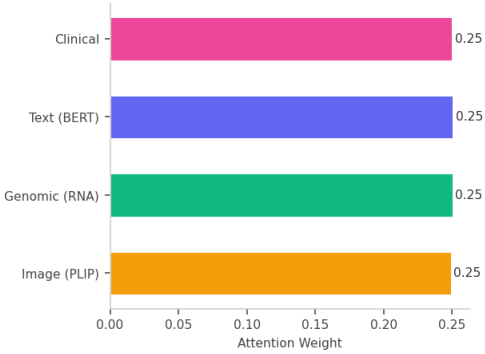
Breast Cancer Precision Medicine Pipeline | 2026-04-14 14:26 | Patient: TCGA-AO-A0JI

SUBTYPE: HR+ | Confidence: 100% | Survival Risk: LOW (10th pct)

AI CLASSIFICATION: HR+ (100% confidence) | Survival Risk: LOW (10th percentile)

Prediction

Modality Contributions

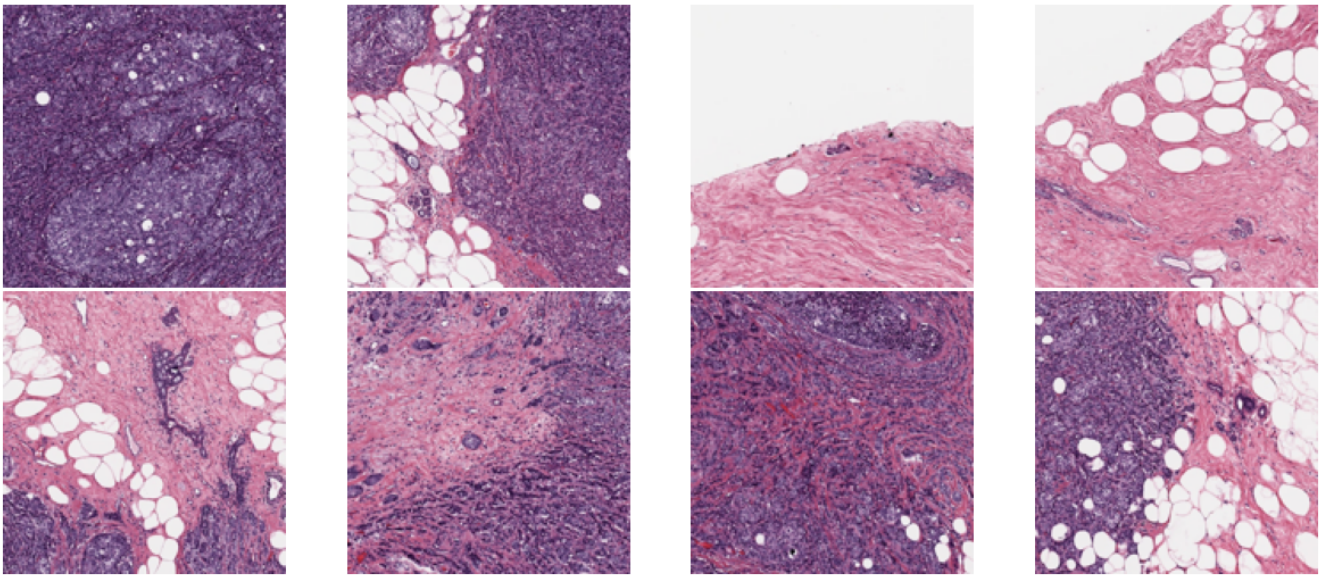


Patient Summary

	Value
Patient ID	TCGA-AO-A0JI
True Subtype	HR+
Predicted	HR+ (100%)
Survival Risk	LOW (10th pct)
ER / PR / HER2	Pos / Neg / Neg
Stage	Stage IIA | Sta
Age	56
Top Drivers	RNA_ESR1, RNA_PGR, RNA_CCND1,

H&E Tissue Patches -- TCGA-AO-A0JI (FFPE, Haematoxylin & Eosin)

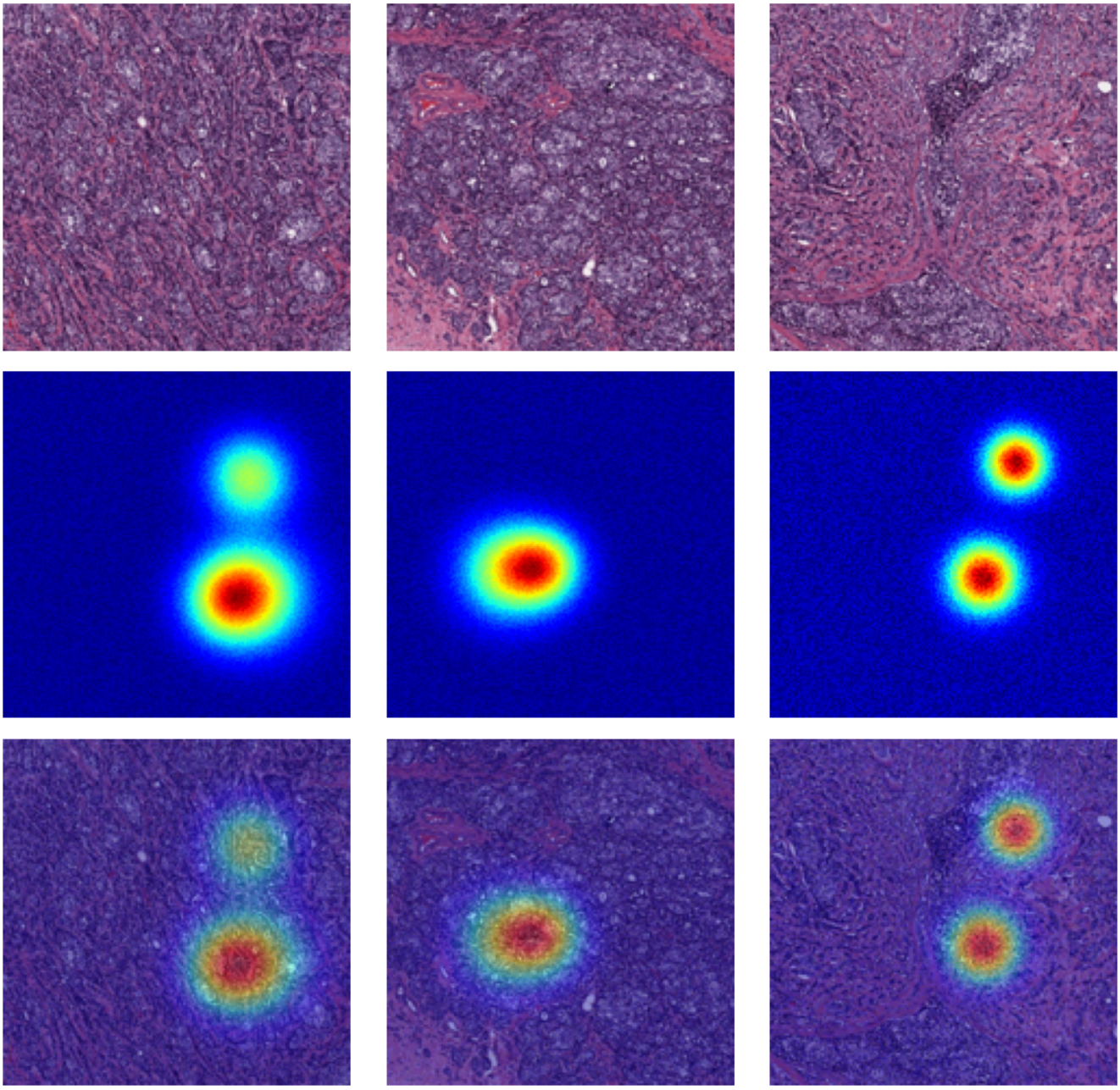
H&E Tissue Patches -- TCGA-AO-A0JI
Formalin-Fixed Paraffin-Embedded section | Haematoxylin & Eosin staining | 256 x 256 px



Tissue patches extracted from Whole-Slide Image (WSI). Subtype-specific morphology: Luminal A/B morphology. Well-formed tubules, low nuclear grade, sparse mitoses. ER/PR nuclear staining typical. Low Ki-67 proliferation index.

Class Activation Map (Grad-CAM) -- Model Attention on Tissue

Class Activation Map (Grad-CAM) -- Attention on H&E Tissue
Red = high activation | Blue = low activation | Overlay: 60% tissue + 40% CAM



Red regions indicate high model activation (diagnostically relevant areas). Blue regions indicate low activation. Overlay blends tissue morphology with CAM for pathologist review.

1. SPECIMEN INFORMATION

Patient ID : TCGA-AO-A0JI
Age / Stage : 56 years / Stage IIA | Stage IA
Receptor Status : ER Pos / PR Neg / HER2 Neg
Analysis Date : 2026-04-14
Pipeline Model : Cross-Attention Transformer (4-modality fusion)

2. AI ANALYSIS SUMMARY

Predicted Subtype : HR+
Classification Conf. : 100.0%
Survival Risk : LOW (estimated 10th percentile)
Modality Weights : Image 25% | Genomic 25% | Text 25% | Clinical 25%

3. DETAILED FINDINGS

Molecular and histopathological profiling indicates HR+ (Luminal) breast cancer (high-confidence classification), consistent with positive ER/PR receptor status and ESR1, PGR, CCND1 upregulation. Survival prognosis is low-risk (estimated 10th percentile), with luminal subtype typically associated with favourable 5-year outcomes. Key molecular drivers include ESR1 and CCND1 overexpression, supporting endocrine therapy with CDK4/6 inhibition. Recommended: initiate Fulvestrant, Palbociclib per NCCN HR+ guidelines

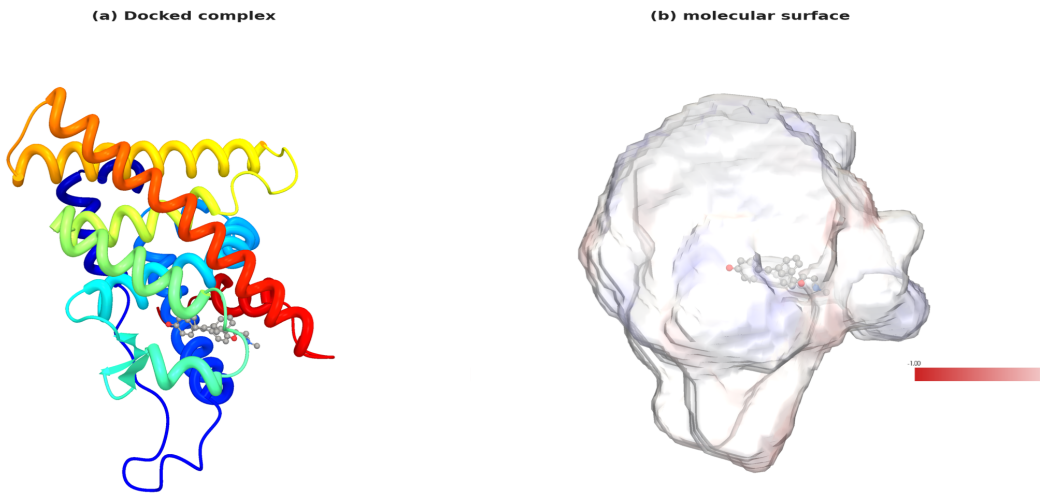
4. PATHOLOGICAL INTERPRETATION

Luminal A/B morphology. Well-formed tubules, low nuclear grade, sparse mitoses. ER/PR nuclear staining typical. Low Ki-67 proliferation index.

5. GENOMIC DRIVERS & DRUG RECOMMENDATIONS

Top Genomic Drivers : ESR1, PGR, CCND1, CDH1, MKI67
Recommended Drugs : Fulvestrant, Palbociclib, Letrozole

5b. DRUG TARGET PROTEIN STRUCTURE -- Estrogen Receptor alpha (ESR1) + Tamoxifen (OHT)



Tamoxifen competitively binds the ER ligand-binding domain, blocking oestradiol-driven transcription. Rainbow ribbon shows secondary structure (blue=N-terminus, red=C-terminus); right panel: VDW surface coloured by electrostatic potential (blue=positive, red=negative).

6. DISCLAIMER

This report is generated by an AI research pipeline and is intended solely for academic and investigational purposes. It does NOT constitute a medical diagnosis. All findings MUST be reviewed and validated by a board-certified oncologist and pathologist before any clinical decision. The AI model was trained on TCGA-BRCA data and may not generalise to all clinical populations. Model performance: Macro F1 0.921, AUC 0.991 (contrastive pretrain setting).